Memory Under Pressure: The Effect of Adrenaline-Driven Treatments

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1 Abstract

Humans often perform differently in certain circumstances, such as states of heightened arousal or stress. This experiment investigates how adrenaline-inducing treatments affect memory performance, measured via recollection ability. Prior to the initiation of our project, we researched current literature pertaining to the relationship between adrenaline and cognitive memory ability with one question in mind: "How do adrenaline inducing treatments affect memory?". Our study utilized a five-by-five Latin Squares design with five replicates, followed by an ANOVA analysis, to investigate this research question. Our goal was to identify if there were any differences in our selected treatments, since our experimental design cannot infer causality. This experiment has the potential to identify which adrenaline-inducing treatments performed differently from the rest, in both the positive and negative direction.

2 Introduction and Literature Review

To select the treatments for our experiment, we reviewed existing literature on potential treatments and their effects on memory. A study from the National Library of Medicine suggested that injections of epinephrine (adrenaline) could positively affect memory ability. The usage of single epinephrine injections can help to prolong memory training capabilities and performance. This is due to the fact that when the brain is put in an emotionally charged state, the body releases stress hormones which support memory formation in the brain region, especially in the amygdala and hippocampus. Another potential treatment that sparked our attention was methamphetamine. Although illegal and widely considered harmful, methamphetamine can have beneficial effects on human memory. According to Neuropsychopharmacology, people with severe and erratic methamphetamine addiction often experience significant cognitive decline. However, several studies have found that low doses of methamphetamine can actually improve memory.

Another drug examined by the National Library of Medicine found an association between cocaine and improved memory. Cocaine is widely recognized to impair brain health and well-being among the scientific community. One study found that increased use over a one-year period was associated with a decline in working memory. However, participants in the study who decreased their cocaine usage showed improvement across all tested cognitive domains.

Physical activity and high-arousal experiences have been shown to elevate adrenaline levels in the short term. An example of this is bungee jumping. Bungee jumping involves leaping from a significant height over canyons while secured to an elastic cord. This simulates the feeling of falling large distances while being safely secured to a harness. The British Psychology Society performed a study where participants had their adrenaline rush triggered via bungee jumping and observed how this affected their cognitive performance. Researchers hypothesized that cognition would be impaired due to high levels of stress. However, the results unexpectedly revealed that memory improved post-activity. The study concluded that acute stress in controlled situations may cause short-term cognitive benefits. Our final treatment that we investigated was a 20 minute High Intensity Interval Training (HIIT) workout. This treatment induces adrenaline through physical exercise that is similar to sports or other physical activities. This seemed like a good comparison to our other treatments as it was a way to observe adrenaline production in participants in a low-stress environment.

After reviewing these five methods of inducing adrenaline, we selected these treatments for our experiment. With additional time, we would have expanded our literature review to investigate even more potentially adrenaline-inducing treatments.

3 Methods

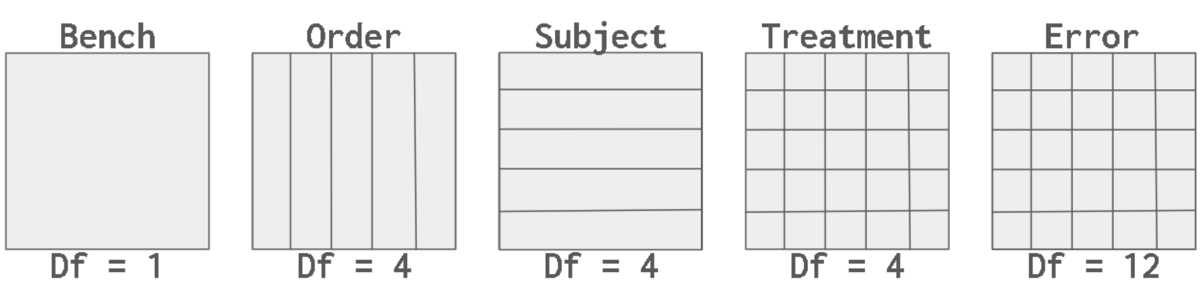
3.1 Design

For our experimental study, we decided on a Latin Squares design. This design method is used to control for two blocking factors alongside a treatment factor. The two blocking factors selected were subjects (testing for subject-to-subject variability) and treatment order. We intended to examine the effects of various adrenaline inducing treatments on our subjects' memory.

In a Latin Squares design, five replicates are generally required to ensure reputable and credible results. Each test is conducted independently. This allows for outcomes to remain specific to each individual Latin Square replicate. Since we had five treatment levels, we assigned five subjects to each Latin Square. As a result, our design involved performing five, "five-by-five", Latin Squares. The parameters for the design are detailed below:

- Rows: 5; Columns: 5; Replicates: 5
- Experimental Units: *The Islands* participants
- Response Variable: Memory Test Score (in seconds)
- Treatments: Bungy Jumping 50m, Adrenaline Injection 10μg, Methamphetamine Injection 50mg, Cocaine Insufflation 50mg, High Intensity Interval Training (HIIT)
- Null Hypothesis: All the treatments have the same or similar effect on a subject's memory
- Alternative Hypothesis: At least one treatment has an effect on memory that was different from the rest of the treatments

The factor diagram is detailed below:



3.2 Participants

The participants of our study are a set of "simulated" people on the website *The Islands*. These participants had to give consent in order to be part of the experiment. Each Latin Square involved arbitrarily selecting five Islanders. While we could have chosen to perform a form of blocking by selecting similar subjects along the lines of sex, age, etc, we did not do so successfully. This does not diminish the findings of our study to a major extent because the Latin Squares design already intends to account for subject-to-subject variability. Each subject received all treatments in a unique, randomized order to control for order effects.

3.3 Procedure

It is vital to ensure proper randomization of treatments in a Latin Squares procedure. Every replicate must have a unique random ordering for the administration of treatments to be considered independent from the prior squares. Every test we produced reflected this requirement by having a different random ordering of treatments.

We used R studio's sample function to produce a pseudo-random ordering of treatments for our first Latin Square. In the table below, an example of the structure of the first Latin Square is shown. Each subject has a unique ordering for receiving each treatment, and each subject received every treatment.

For the subsequent replicates, each was generated by reordering the rows of the first replicate in four distinct ways to create five total replicates.

Subject	Treatment 1	Treatment 2	Treatment 3	Treatment 4	Treatment 5
1	E	C	D	B	A
2	D	B	C	A	E
3	B	E	A	D	C
4	A	D	E	C	B
5	C	A	B	E	D

Replicate 1 Latin Square Structure Example

Each subject was tested once per day to maintain consistency throughout the experiment and to allow the effects of the prior treatment to wear off before administering the next treatment. For each treatment, we conducted preliminary research to determine the adequate time between treatment and test. This required waiting periods in some cases (Cocaine Insufflation), and testing immediately in others (HIIT). The exact procedures we used are detailed below:

- (A) Bungee Jumping 50m - wait 5 minutes - memory test
- (B) Adrenaline Injection 10μg - wait 5 minutes - memory test
- (C) Methamphetamine Injection 50mg - memory test
- (D) Cocaine Insufflation 50mg - wait 15 minutes - memory test
- (E) HIIT 20 minutes - memory test

4 Data Analysis

4.1 Type of Statistical Analysis

To evaluate the effects of various adrenaline-inducing treatments on memory performance, we employed a Latin Squares Design with five replicates while accounting for two sources of nuisance variability: the identity of the subject and the treatment order. The response variable was memory, examined via a memory game test performance (in seconds) after each treatment.

For the analysis, we constructed separate ANOVA models for each Latin Square replicate to assess within-replicate variability and analyze each square independently. Post-hoc pairwise comparisons using Tukey's HSD tests were employed as needed to identify specific differences between treatments on models that were statistically significant. Model diagnostics were produced and analyzed to verify the model assumptions of normality, constant variance, and linearity were met.

4.2 Sample Size Determination

In Latin Squares designs, there is no standardized method for determining sample size. Latin Squares requires substantially more observations per replicate than many standard designs due to every subject receiving every treatment. This causes reputability of a Latin Squares design to be measured in its reproducibility. In general, at least five replicates are needed in order to achieve a benchmark level of credibility, with each replicate improving the design.

Each Latin Square replicate included five participants, with each being exposed to all five treatments in a randomized order. To meet the benchmark, we performed five separate Latin Squares which amounted to a total of 125 observations. The Latin Squares structure ensures that every treatment appears exactly once in each row and column per replicate to control for order and subject variability.

The diagnostic plots for some of our Latin Squares confirmed that the conditions of constant variance, linearity, and normality of errors were met. However, a few replicates had diagnostic plots that violated some model assumptions. We addressed this by performing transformations on our response variable to better satisfy model assumptions (more details on model validity in section 5.3).

4.3 Causality

An understanding of the limitations of causality is required when discussing Latin Squares. Utilizing a Latin Squares design does not allow for conclusions about any specific treatment causing an outcome in the response variable because there is no control group. The goal of this design is to compare an array of treatments to each other with regards to a response variable. Hence, we are looking for differences between our treatments pertaining to their respective effect on memory rather than a specific cause of a boosted or hindered memory. While we can conclude one treatment outperformed or underperformed others in terms of memory, we cannot infer its definitive causal relationship with memory overall.

5 Results

5.1 ANOVA Results

We performed an ANOVA test separately for each of the five Latin Squares. The null hypothesis stated that there was no difference across treatments in mean memory response time, while the alternative hypothesis posited that at least one treatment differed. The AOV model was structured with an additive approach since the Latin Squares design does not allow for the usage or consideration of interaction. The primary focus was on the Treatment variable. Our goal was to determine whether certain treatments led to significantly better memory response times than others. We used a significance level of $\alpha = .05$. Below are the summaries of our AOV model:

Source	**Df**	**Sum Sq**	**Mean Sq**	**F value**	**Pr**$(> F)$
Treatment	4	175.8	43.9	1.59	.239987
Subject	4	1326.7	331.7	12	.000371 ***
Order	4	97.1	24.3	.879	.505122
Residuals	12	331.7	27.6		

Replicate 1: ANOVA Table for Latin Square Design

Source	**Df**	**Sum Sq**	**Mean Sq**	**F value**	**Pr**$(> F)$
Treatment	4	216	54	2.844	.0718
Subject	4	6449	1612	84.973	1.07e-08 ***
Order	4	60	14.9	.787	.5554
Residuals	12	228	19		

Replicate 2: ANOVA Table for Latin Square Design

Source	**Df**	**Sum Sq**	**Mean Sq**	**F value**	**Pr**$(> F)$
Treatment	4	3.995e-08	9.990e-09	5.853	.00752**
Subject	4	1.795e-07	4.486e-08	26.294	7.37e-06
Order	4	1.608e-08	4.020e-09	2.357	.11232
Residuals	12	2.047e-08	1.710e-09		

Replicate 3: ANOVA Table for Latin Square Design ($\frac{1}{Y^2}$ Transformation)

Source	**Df**	**Sum Sq**	**Mean Sq**	**F value**	**Pr**$(> F)$
Treatment	4	69	17.4	.723	.593
Subject	4	6716	1678.9	69.923	3.27e-08 ***
Order	4	133	33.3	1.385	.297
Residuals	12	288	24		

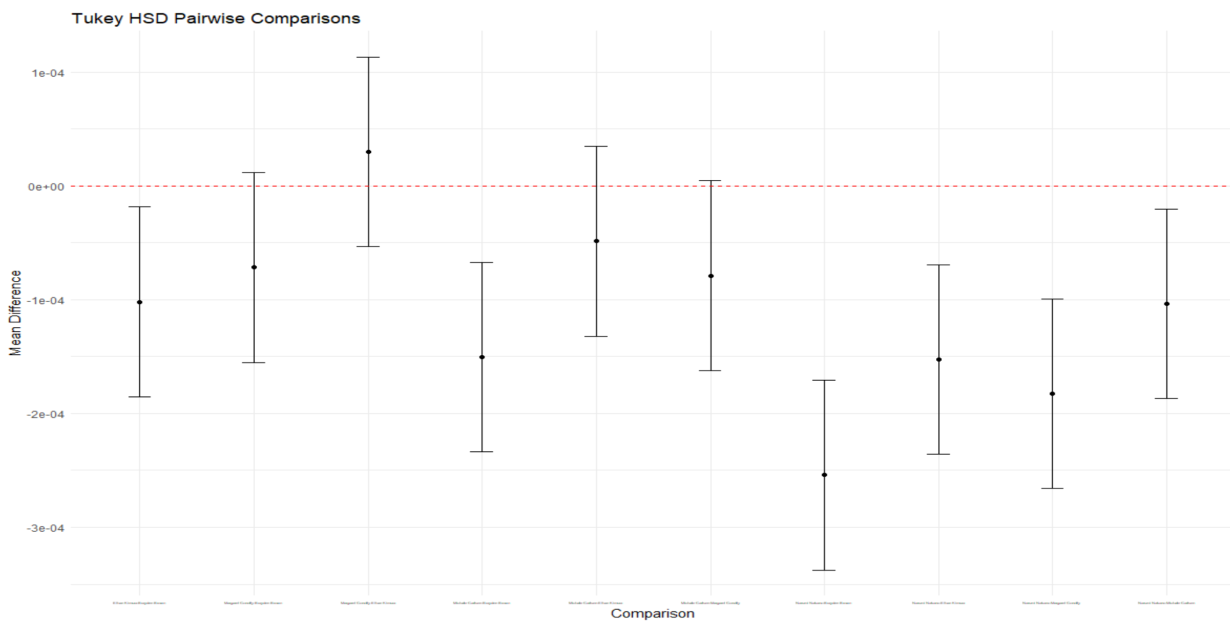
Replicate 4: ANOVA Table for Latin Square Design

Source	Df	Sum Sq	Mean Sq	F value	Pr(> F)
Treatment	4	156612	391653	.889	.500
Subject	4	105356487	26339122	59.756	8.01e-08 ***
Order	4	791742	197935	.449	.771
Residuals	12	5289366	440780		

Replicate 5: ANOVA Table for Latin Square Design (Y^2 Transformation)

5.2 Post-Hoc

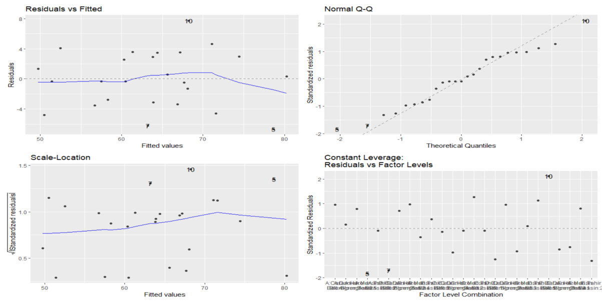
We performed a single post-hoc test for *Latin Square 3* since the p-value for the Treatment variable in the ANOVA table was significant. The significant p-value for Treatment in Square Three indicates that the type of treatment had an impact on the memory game test scores within Square Three. We did not perform post-hoc tests for the other replicates because the Treatment variable was not significant there.



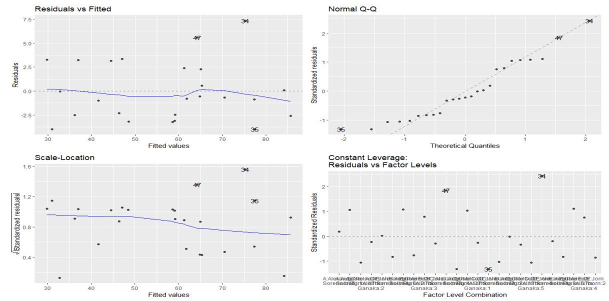
Replicate 3

5.3 Diagnostic Plots

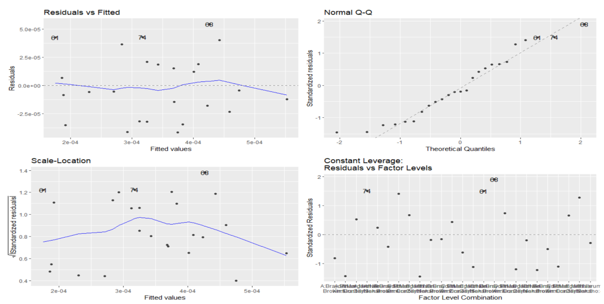
The diagnostic plots for each Latin Squares are presented below:



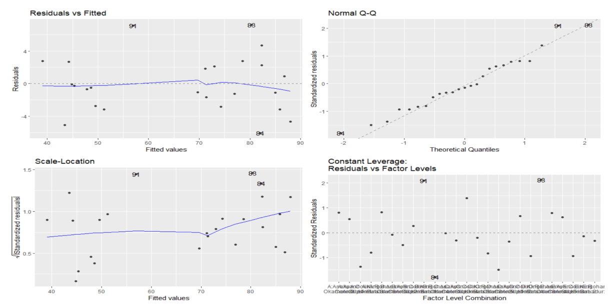
Replicate 1



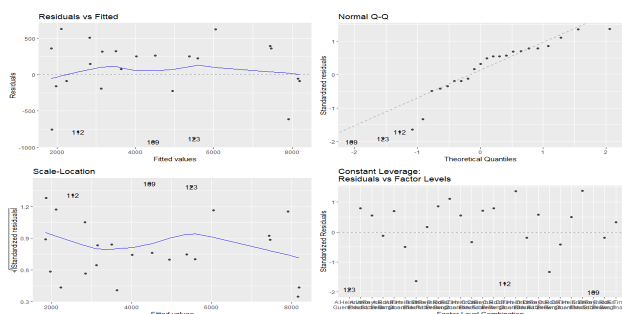
Replicate 2



Replicate 3: $1/Y^2$ Transform



Replicate 4



Replicate 5: Y^2 Transform

Diagnostic plots for Latin Square replicates 1, 2, and 4 showed a reasonably constant variance and an approximate normality of the residuals. Based on the residuals vs. fitted value plots for these three Latin Squares, there is a random scatter of points around zero with no noticeable curvature or funnel shapes, indicating homoscedasticity and linearity. The Q-Q Plots for these three squares demonstrated that the standardized residuals closely followed the dotted line, suggesting that the normality condition was reasonably met. The diagnostics for Latin Square replicates 1,2, and 4 suggested that the underlying assumptions of independence, constant variance, and normality of residuals for the ANOVA models were met, confirming model validity.

Initial diagnostics analyses for replicates 3 and 5 indicated violations of the model assumptions, primarily concerning constant variance. The validity of the model is crucial for credibility of results because an invalid model can compromise all inferential tools. Therefore, we applied transformations to the response variables of these two Latin squares to ensure that they follow satisfactory model validity.

In order to decide on the appropriate transform for these two models, we utilized RStudio and its *boxcox* and *powerTransform* functions to identify some ideal transformations that could be applied to our models to potentially improve their validity. We settled on a transformation of $\frac{1}{Y^2}$ for our third Latin Square and a transformation of Y^2 for our fifth Latin Square, where Y refers to our response variable (memory). Once these transformations were applied, we reproduced the diagnostic plots (presented above). The conditions for constant variance, normality, and linearity were satisfied much better after the transformations. Consequentially, we kept these transformations for Latin Square replicates three and five to ensure that our model conditions are satisfied.

6 Conclusion

This study aimed to investigate the effects of various adrenaline inducing treatments affect memory. We utilized a Latin Squares design to administer each of the five treatments to five subjects in unique sequences across five replicates. Our ANOVA analysis focused on investigating whether any of these treatments varied significantly from one another in regards to our memory response variable. The design controlled for two nuisance factors: treatment order and subject-to-subject variability, while primarily intending to investigate treatment effects.

The null hypothesis of our study is that all the treatments have the same or similar effect on a subject's memory. The alternative hypothesis is that at least one treatment has an effect on memory that was different from the rest of the treatments. Subject memory was observed through a memory game, with ANOVA tests performed to analyze the mean response times. It is important to note that these means represent marginal means calculated across subjects within each Latin Square, rather than within-subjects means. This does not conflate or confound with subject-to-subject variability because the averages are expected to be influenced similarly by each subject. If one subject typically has faster memory, their contribution to each marginal treatment mean will be correspondingly lower because they already have a fast base memory. Likewise, a subject with slower memory contributes higher values to the marginal treatment means on a relatively consistent level.

With this clarification, comparisons of marginal means for each treatment can be made. This is done all at once in the ANOVA analysis, but we would need to perform post-hoc testing, such as Tukey's HSD test, to identify which specific pairwise comparisons of treatment means differed. This is because the ANOVA process corresponds to the earlier hypotheses stating that if the at least one treatment mean differs, the null is rejected, and otherwise we fail to reject the null. The ANOVA test follows the standard F-test procedure which does not identify individual treatment differences but rather compares them all at once.

Taking a look at our ANOVA tables in section 5.1, there is one for each Latin Square replicate. To understand significance across treatment means, we are only interested in the first row of each ANOVA table (Treatment). This row identifies if any differences were present across treatment means (p-value < 0.05). It is important to understand why the other two rows are not of interest here. Earlier, we mentioned that we have two nuisance factors, order and subject variability. These two nuisance factors correspond to the rows in the table and are being tested to an extent. Therefore, we can observe their effects and significance, but this does not influence or help us better understand our initial research question.

In all our ANOVA tables, there was not a single time where the Order variable produced a significant p-value. This means that order of treatment administration is both a nuisance factor and not statistically significant in impact on our response variable across the board. In all of our ANOVA tables, it is noticeable that the Subject variable always produced a statistically significant p-value. This signals to us that subject-to-subject variability plays a significant role in predicting or understanding our response variable. This makes sense for a couple reasons. First, we already might anticipate, that different subjects will have different reactions when exposed to different treatments. Second, we mentioned earlier that we did not properly block in this study. What this means is that we did not have a general criterion for mitigating subject-to-subject variability holistically. Ideally, we would have selected an age range, a gender, and a couple other defining characteristics across islanders to block for. We would have executed this blocking by ensuring that all the selected subjects in each Latin Square met these criteria consistently. This would then reduce subject-to-subject variability because subjects were already identified and filtered along factors that could play a potential role in affecting their memory capabilities. This suggests our Subject variable is not noteworthy even though it is significant in all our tables, in addition to also being a nuisance factor.

To discuss the Treatment variable, we can observe the corresponding p-values across our five ANOVA tables for each Latin Squares replicate. In replicates one, two, four, and five, the produced p-value in the Treatment row was above our intended significance level of $\alpha = 0.05$. Thus, there was not a statistically significant difference in marginal treatment means across subjects with regards to the response variable of memory in four out of five replicates. In Latin Square three, we received a statistically significant p-value for the Treatment variable. Given that this didn't happen in other cases, it is somewhat perplexing.

We performed a Tukey Post-Hoc test of pairwise comparison for Latin Square three due to the presence of a statistically significant p-value in the Treatment variable. In section 5.2, most (7/10) of the confidence intervals produced for pairwise comparisons of mean differences across treatments did not include 0. This means that differences between treatment means were present in many cases, which might suggest why the p-value produced was statistically significant.

Despite the significance in square three, it seems that in most cases our Treatment variable was not significant. As a result, we fail to reject the null that there is not a significant difference between these five adrenaline inducing treatments (Cocaine, Methamphetamine, Adrenaline, HIIT, Bungee Jumping) when it comes to their effects on memory. The implications and general relevancy of this study are limited because we only looked at five treatments. It is still interesting that none of these treatments were statistically significant in their effects on memory in comparison to the others because they are so different. We cannot infer about causality due to our experimental design, but perhaps we saw the results we did because all treatments activated similar levels of adrenaline in patients or because adrenaline doesn't have an effect on memory. However, these are all things we cannot conclude from our study because of our Latin Squares design. Our conclusion is that these treatments did not differ significantly across our replicates in regards to memory performance of subjects on the Islands.

7 References

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